

46 YEARS OF EXCELLENCE | SRAMAYEVA JAYATHE | NARAYANA IIT - JEE ACADEMY - INDIA

Sec: SR.IIT_*CO-SC(MODEL-A& APEX) **GTM-22** **Date: 21.01.2025**
Time: 3.00Hrs. **JEE MAIN MODEL** **Max. Marks: 300**

PHYSICS MAX.MARKS: 100

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

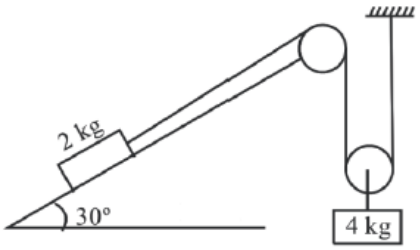
1. Match List-I with List-II

	List-I		List-II
a.	Coefficient of viscosity	I.	$[ML^2T^{-2}]$
b.	Surface Tension	II.	$[ML^2T^{-1}]$
c.	Angular momentum	III.	$[ML^{-1}T^{-1}]$
d.	Rotational kinetic energy	IV.	$[ML^0T^{-2}]$

A) $a-II, b-I, c-IV, d-III$ B) $a-I, b-II, c-III, d-IV$

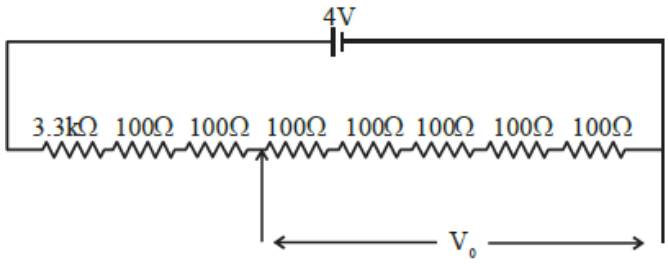
C) $a-III, b-IV, c-II, d-I$ D) $a-IV, b-III, c-II, d-I$

2. All surfaces shown in figure are assumed to be frictionless and the pulleys and the string are light. The acceleration of the block of mass 2 kg is:



A) g B) $\frac{g}{3}$ C) $\frac{g}{2}$ D) $\frac{g}{4}$

3. A potential divider circuit is shown in figure. The output voltage V_0 is



A) $4V$ B) $2mV$ C) $0.5V$ D) $12mV$

4. Young's modules of material of a wire of length 'L' and cross-sectional area A is Y. If the length of the wire is doubled and cross-sectional area is halved then Young's modules will be

A) $\frac{Y}{4}$ B) $4Y$ C) Y D) $2Y$

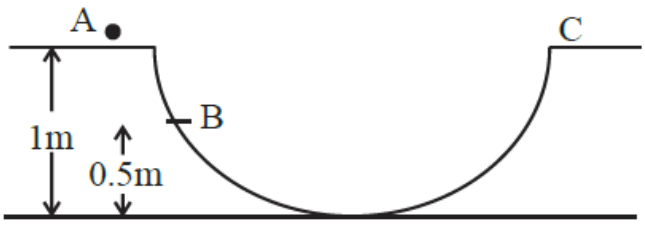
5. The work function of a substance is 3.0 eV. The longest wavelength of light that can cause the emission of photoelectrons from this substance is approximately:

A) 215 nm B) 414 nm C) 400 nm D) 200 nm

6. The ratio of the magnitude of the kinetic energy to the potential energy of an electron in the 5th excited state of a hydrogen atom is:

A) 4 B) $\frac{1}{4}$ C) $\frac{1}{2}$ D)1

7. A particle is placed at the point A of a frictionless track ABC as shown in figure. It is gently pushed toward right. The speed of the particle when it reaches the point B is: (Take $g = 10 m/s^2$)



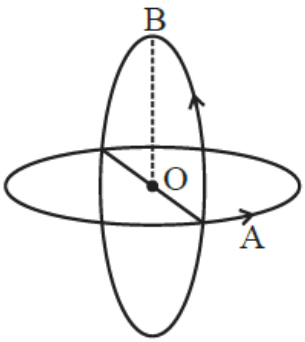
A) $20\ m/s$ B) $\sqrt{10}\ m/s$ C) $2\sqrt{10}\ m/s$ D) $10\ m/s$

8. The electric field of an electromagnetic wave in free space is represented as $\vec{E} = E_0 \cos(\omega t - kz)\hat{i}$. The corresponding magnetic induction vector will be:

A) $\vec{B} = E_0 C \cos(\omega t - kz)\hat{j}$ B) $\vec{B} = \frac{E_0}{C}\cos(\omega t - kz)\hat{j}$

C) $\vec{B} = E_0 C \cos(\omega t + kz)\hat{j}$ D) $\vec{B} = \frac{E_0}{C}\cos(\omega t + kz)\hat{j}$

9. Two insulated circular loop A and B radius 'a' carrying a current of 'I' in the anti clockwise direction as shown in figure. The magnitude of the magnetic induction at the centre will be:



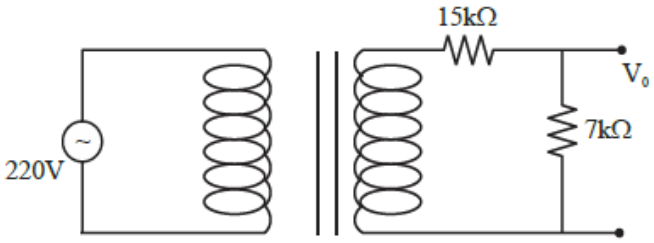
A) $\frac{\sqrt{2}\mu_0 I}{a}$ B) $\frac{\mu_0 I}{2a}$ C) $\frac{\mu_0 I}{\sqrt{2}a}$ D) $\frac{2\mu_0 I}{a}$

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10. The diffraction pattern of a light of wavelength 400 nm diffracting from a slit of width 0.2 mm is focused on the focal plane of a convex lens of focal length 100 cm. The width of the 1st secondary maxima will be:

A) 2 mm B) 2 cm C) 0.02 mm D) 0.2 mm

11. Primary coil of a transformer is connected to 220 V ac. Primary and secondary turns of the transforms are 100 and 10 respectively. Secondary coil of transformer is connected to two series resistance shown in shown in figure. The output voltage (V_0) is:



A) $7\,V$ B) $15\,V$ C) $44\,V$ D) $22\,V$

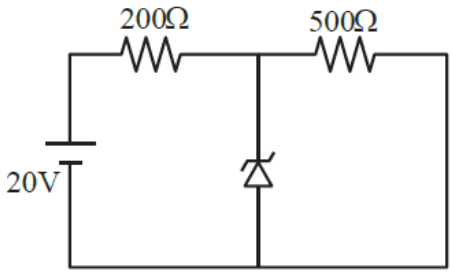
12. The gravitational potential at a point above the surface of earth is $-5.12\times10^{7}\ J/kg$ and the acceleration due to gravity at the point is $6.4\,m/s^2$. Assume that the mean radius of earth to be $6400\ km$. The height of this point above the earth's surface is:

A) 1600 *km* B) 540 *km* C) 1200 *km* D) 1000 *km*

13. An electric toaster has resistance of 60Ω at room temperature $(27°C)$. The toaster is connected to a 220 V supply. If the current flowing through it reaches 2.75 A, the temperature attained by toaster is around: (if $\alpha = 2\times10^{-4}/°C$)

A) $694°C$ B) $1235°C$ C) $1694°C$ D) $1667°C$

14. A Zener diode of breakdown voltage 10V is used as a voltage regulator as shown in the figure. The current through the Zener diode is



A) 50 mA B) 0 C) 30 mA D)20 mA

15. The distance between object and its two times magnified real image as produced by a convex lens is 45 cm. The focal length of the lens used is ……..cm.

A) 10 cm B) 30 cm C) 40 cm D) 50 cm

16. The electrostatic potential due to an electric dipole at a distance ‘ r ’ varies as:

A) r B) $\frac{1}{r^2}$ C) $\frac{1}{r^3}$ D) $\frac{1}{r}$

17. A spherical body of mass 100 g is dropped from a height of 10 m from the ground. After hitting the ground, the body rebounds to a height of 5 m. The impulse of force imparted by the ground to the body is given by: (Given: $g = 9.8 kg\ ms^{-2}$)

A) $4.32\ kg\ ms^{-1}$ B) $43.2\ kg\ ms^{-1}$ C) $23.9\ kg\ ms^{-1}$ D) $2.39\ kg\ ms^{-1}$

18. A particle of mass m projected with a velocity ‘u’ making an angle of $30°$ with the horizontal. The magnitude of angular momentum of the projectile about the point of projection when the particle is at its maximum height h is:

A) $\frac{\sqrt{3}}{16}\frac{mu^3}{g}$ B) $\frac{\sqrt{3}}{2}\frac{mu^2}{g}$ C) $\frac{mu^3}{\sqrt{2}g}$ D) Zero

19. At which temperature the r.m.s. velocity of a hydrogen molecule equal to that of an oxygen molecule at $47°C$?

A) $80\ K$ B) $-73\ K$ C) $4\ K$ D) $20\ K$

20. A series L, R circuit connected with an ac source $E=(25\sin 1000t)V$ has a power factor of $\frac{1}{\sqrt{2}}$. If the source of emf is changed to $E=(20\sin 2000t)V$, the new power factor of the circuit will be:

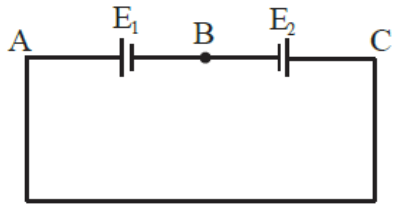
A) $\frac{1}{\sqrt{2}}$ B) $\frac{1}{\sqrt{3}}$ C) $\frac{1}{\sqrt{5}}$ D) $\frac{1}{\sqrt{7}}$

SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals , **Round off to the nearest Integer only.** **Have to Answer all 5 questions** and question will be evaluated according to the following marking scheme: Marking Scheme: +4 for correct answer, 0 if not attempted and –1 if not correct.

21. The horizontal component of earth's magnetic field at a place is $3.5\times10^{-5}T$. A very long straight conductor carrying current of $\sqrt{2}A$ in the direction from South east to North west is placed. The force per unit length experienced by the conductor is _____$\times10^{-6}N/m$.

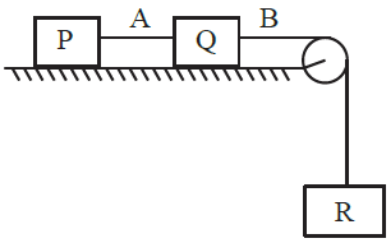
22. Two cells are connected in opposition as shown. Cell E_1 is of $8V$ emf and 2Ω internal resistance; the cell E_2 is of $2V$ emf and 4Ω internal resistance. The terminal potential difference of cell E_2 is:



23. A electron of hydrogen atom on an excited state is having energy $E_n=-0.85\,eV$. The maximum number of allowed transitions to lower energy level is ……

24. Each of three blocks P, Q and R shown in figure has a mass of 3 kg. Each of the wire A and B has cross-sectional area $0.005\,cm^2$ and Young's modulus $2\times10^{11}\,Nm^{-2}$. Neglecting friction, the longitudinal strain on wire B is ……..$\times10^{-4}$.
(Take: $g=10\,m/s^2$)

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25. A capacitor of capacitance C and potential V has energy E. It is connected to another capacitor of capacitance $2C$ and potential $2V$. Then the loss of energy is $\frac{x}{3}E$, Where x is _____

CHEMISTRY MAX.MARKS: 100

SECTION – I
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Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

26. Given below are two statements:

Statement-I: The gas liberated on warming a salt with dil H_2SO_4, turns a piece of paper dipped in lead acetate into black, it is a confirmatory test for sulphide ion.

Statement-II: In statement-I the colour of paper turns black because of formation of lead sulphite.

In the light of the above statements, choose the most appropriate answer from the options given below:

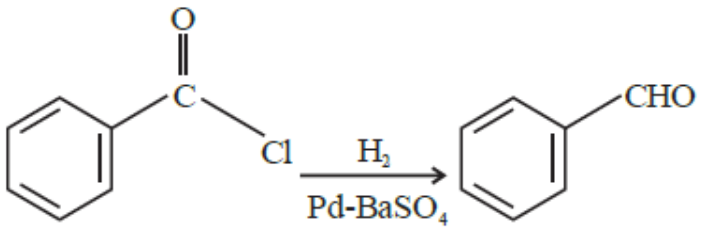
A) Both Statement-I and Statement-II are false

B) Statement-I is false but Statement-II is true

C) Statement-I is true but Statement-II is false

D) Both Statement-I and Statement-II are true.

27.



This reduction reaction is known as:

A) Rosenmund reduction B) Wolff-Kishner reduction

C) Stephen reduction D) Etard reduction

28. Sugar which does not give reddish brown precipitate with Fehling‟s reagent is:

A) Sucrose B) Lactose C) Glucose D) Maltose

29. Given below are the two statements: one is labeled as Assertion (A) and the other is labeled as Reason (R).

Assertion (A): There is a considerable increase in covalent radius from N to P. However from As to Bi only a small increase in covalent radius is observed.

Reason (R): covalent and ionic radii in a particular oxidation state increases down the group.

In the light of the above statement, choose the most appropriate answer from the options given below:

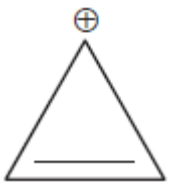
A) (A) is false but (R) is true

B) Both (A) and (R) are true but (R) is not the correct explanation of (A)

C) (A) is true but (R) is false

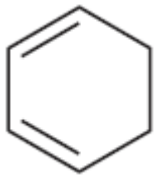
D) Both (A) and (R) are true and (R) is the correct explanation of (A)

30. Which of the following molecule/species is most stable?

A) 

B) 

C) 

D) 

31. Diamagnetic Lanthanoid ions are:

A) Nd^{3+} and Eu^{3+} B) La^{3+} and Ce^{4+}

C) Nd^{3+} and Ce^{4+} D) Lu^{3+} and Eu^{3+}

32. Aluminium chloride in acidified aqueous solution forms an ion having geometry

A) Octahedral B) Square Planar

C) Tetrahedral D) Trigonal bipyramidal

33. Given below are two statements:

Statement-I: The orbitals having same energy are called as degenerate orbitals.

Statement-II: In hydrogen atom, 3p and 3d orbitals are not degenerate orbitals.

In the light of the above statements, choose the **most appropriate** answer from the options given

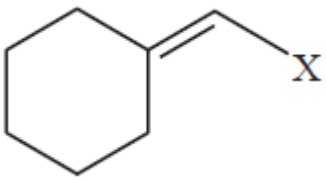
A) Statement-I is true but Statement-II is false

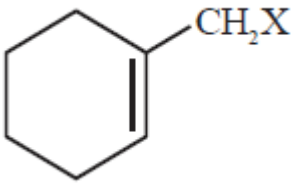
B) Both Statement-I and Statement-II are true

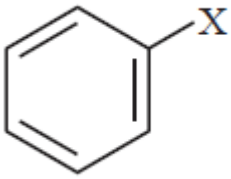
C) Both Statement-I and Statement-II are false

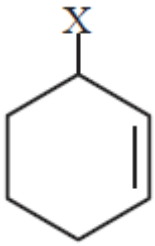
D) Statement-I is false but Statement-II is true

34. Example of vinylic halide is

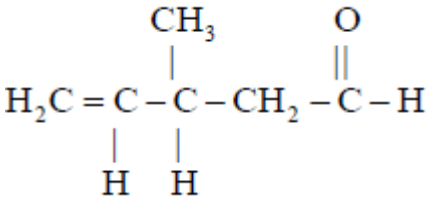
A) 

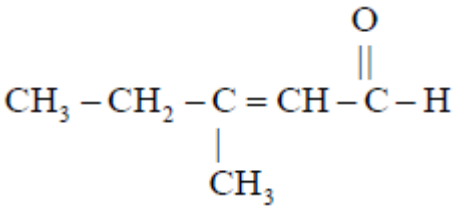
B) 

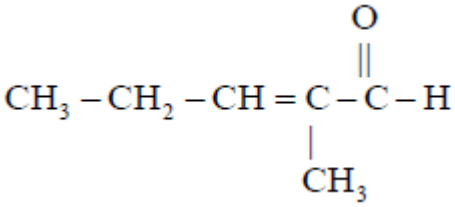
C) 

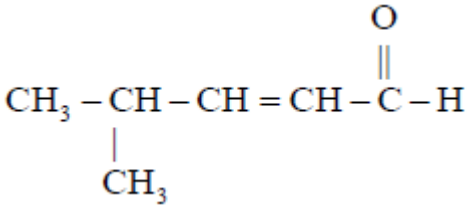
D) 

35. Structure of 4-Methylpent-2-enal is

A) 

B) 

C) 

D) 

36. Match List-I with List-II

	List-I **Molecule**		**List-II** **Shape**
a)	BrF_5	I)	T-shape
b)	H_2O	II)	See saw
c)	ClF_3	III)	Bent
d)	SF_4	IV)	Square pyramidal

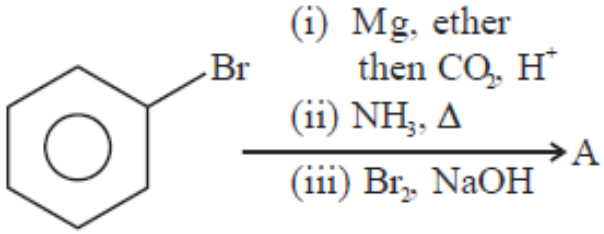
A) (a)-I, (b)-II, (c)-IV, (d)-III

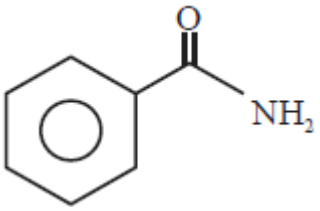
B) (a) –II, (b)-I, (c)-III, (d)-IV

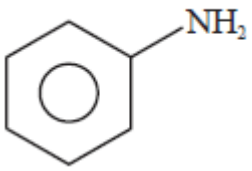
C) (a)-III, (b)-IV, (c)-I, (d)-II

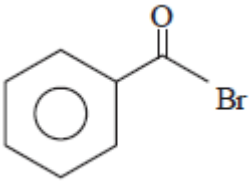
D) (a)-IV, (b)-III, (c)-I, (d)-II

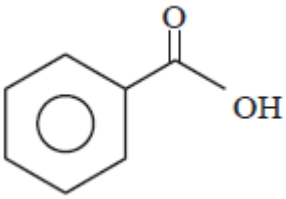
37. The final product A, formed in the following multistep reaction sequence is:



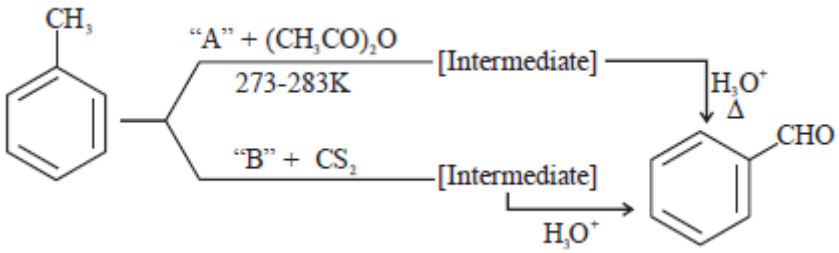
A) 

B) 

C) 

D) 

38. In the given reactions identify the reagent A and reagent B



A) $A-CrO_3$ $B-CrO_3$ B) $A-CrO_3$ $B-CrO_2Cl_2$

C) $A-CrO_2Cl_2$ $B-CrO_2Cl_2$ D) $A-CrO_2Cl_2$ $B-CrO_3$

39. Given below are two statement one is labeled as **Assertion (A)** and the other is labeled as **Reason (R).**

Assertion (A): $CH_2 = CH - CH_2 - Cl$ is an example of allyl halide

Reason (R): Allyl halides are the compounds in which the halogen atom is attached to sp^2 hybridised carbon atom.

In the light of the two above statements, choose the **most appropriate** answer from the options given below:

A) (A) is true but (R) is false

B) Both (A) and (R) are true but (R) is not the correct explanation of (A)

C) (A) is false but (R) is true

D) Both (A) and (R) are true and (R) is the correct explanation of (A)

40. What happens to freezing point of benzene when small quantity of naphthalene is added to benzene?

A) Increases B)Remains unchanged

C) First decreases and then increases D) Decreases

41. Match List-I with List-II

	List-I Species		**List-II Electronic distribution**
a.	Cr^{+2}	I.	$3d^8$
b.	Mn^{+}	II.	$3d^3 4s^1$
c.	Ni^{+2}	III.	$3d^4$
d	V^{+}	IV.	$3d^5 4s^1$

Choose the correct answer from the options given below:

A) $a-I, b-II, c-III, d-IV$ B) $a-III, b-IV, c-I, d-II$

C) $a-IV, b-III, c-I, d-II$ D) $a-II, b-I, c-IV, d-III$

42. Compound A formed in the following reaction reacts with B gives the product C. Find out A and B.

$$CH_3-C\equiv CH + Na \rightarrow A \xrightarrow{B} \underset{(C)}{CH_3-C\equiv C-CH_2-\underset{\displaystyle CH_3}{\underset{|}{CH_2}}} + NaBr$$

A) $A=CH_3-C\equiv \bar{C}N\overset{+}{a}$, $B = CH_3-CH_2-CH_2-Br$

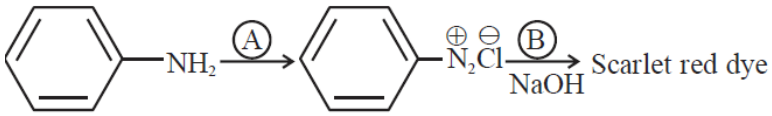
B) $A=CH_3-CH=CH_2$, $B = CH_3-CH_2-CH_2-Br$

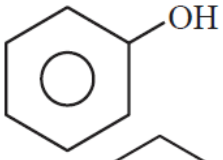
C) $A = CH_3-CH_2-CH_3$, $B = CH_3-C\equiv CH$

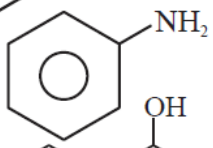
D) $A = CH_3-C\equiv \bar{C}N\overset{+}{a}$, $B = CH_3-CH_2-CH_3$

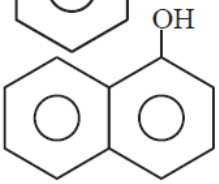
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43. Following is a confirmatory test for aromatic primary amines. Identify reagent (A) and (B)

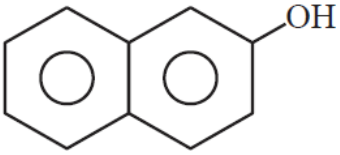


A) A = HNO_3/H_2SO_4; B = 

B) A= $NaNO_2$ + HCl, 0 – 5°C; B = 

C) A=$NaNO_2$ + HCl, 0 – 5°C; B = 

D) A = $NaNO_2$ + HCl, 0 – 5°C;

B = 

44. The Lassiagne's extract is boiled with dil HNO_3 before testing for halogens because,

A) $AgCN$ is soluble in HNO_3

B) Silver halides are soluble in HNO_3

C) Ag_2S is soluble in HNO_3

D) Na_2S and $NaCN$ are decomposed by HNO_3

45. Choose the correct statements from the following.

a) Ethane-1,2-diamine is a chelating ligand.

b) Metallic aluminium is produced by electrolysis of aluminium oxide in presence of cryolite.

c) Cyanide ion is used as ligand for leaching of silver

d) Phosphine act as a ligand in Wilkinson catalyst.

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e) The stability constants of Ca^{2+} and Mg^{2+} are similar with EDTA complexes.

Choose the correct answer from the options given below.

A) b, c, e only B) c, d, e only C) a, b, c only D) a, d, e only

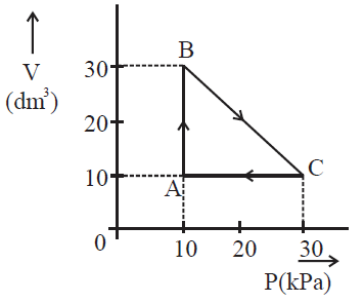
SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals , **Round off to the nearest Integer only.** **Have to Answer all 5 questions** and question will be evaluated according to the following marking scheme: Marking Scheme: +4 for correct answer, 0 if not attempted and –1 if not correct.

46. The rate of first order reaction is $0.04 mol\, L^{-1}s^{-1}$ at 10 minutes and $0.03\, mol\ L^{-1}s^{-1}$ at 20 minutes after initiation. Half life of the reaction is _____ minutes. (Given $\log 2 = 0.3010, \log 3 = 0.4771$)

47. At $25°C$, the pH at which $Mg(OH)_2\left[K_{sp} = 1\times10^{-11}\right]$ begins to precipitate from a solution containing 0.10 M Mg^{2+} ions is _____

48.



An ideal gas undergoes a cyclic transformation starting from the point A and coming back to the same point by tracing the path A → B → C → A as shown in the diagram. The total work done in the process is _____ J.

49. If IUPAC name of an element is "Unununnium" then the element belongs to nth group of periodic table. The value of n is_____

50. The total number of molecular orbitals formed from 2s and 2p atomic orbitals of a diatomic molecule

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MATHS **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

51. A line passing through the point $A(9,0)$ makes an angle of $30°$ with the positive direction of $x-$ axis. If this line is rotated about A through an angle of $15°$ in the clockwise direction, then its equation in the new position is

A) $\frac{y}{\sqrt{3}-2}+x=9$ B) $\frac{x}{\sqrt{3}-2}+y=9$

C) $\frac{x}{\sqrt{3}+2}+y=9$ D) $\frac{y}{\sqrt{3}+2}+x=9$

52. Let S_n denote the sum of first n terms of an arithmetic progression. If $S_{20}=790$ and $S_{10}=145$, then $S_{15}-S_5$ is

A) 395 B) 390 C) 405 D) 410

53. If $z=x+iy, xy\neq 0$ satisfies the equation $z^2+i\bar{z}=0$, then $\left|z^2\right|$ is equal to:

A) 9 B) 1 C) 4 D) $\frac{1}{4}$

54. Let $\vec{a}=a_1\hat{i}+a_2\hat{j}+a_3\hat{k}$ and $\vec{b}=b_1\hat{i}+b_2\hat{j}+b_3\hat{k}$ be two vectors such that $\left|\vec{a}\right|=1; \vec{a}.\vec{b}=2$ and $\left|\vec{b}\right|=4$. If $\vec{c}=2\left(\vec{a}\times\vec{b}\right)-3\vec{b}$, then the angle between $\vec{b}$ and $\vec{c}$ is equal to:

A) $\cos^{-1}\left(\frac{2}{\sqrt{3}}\right)$ B) $\cos^{-1}\left(-\frac{1}{\sqrt{3}}\right)$ C) $\cos^{-1}\left(-\frac{\sqrt{3}}{2}\right)$ D) $\cos^{-1}\left(\frac{2}{3}\right)$

55. The maximum area of a triangle whose one vertex is at $(0,0)$ and the other two vertices lie on the curve $y=-2x^2+54$ at points (x,y) and $(-x,y)$ where $y>0$ is:

A) 88 B) 122 C) 92 D) 108

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56. The value of $\lim\limits_{n\to\infty}\sum\limits_{k=1}^{n}\dfrac{n^3}{\left(n^2+k^2\right)\left(n^2+3k^2\right)}$ is:

A) $\dfrac{\left(2\sqrt{3}+3\right)\pi}{24}$ B) $\dfrac{13\pi}{8\left(4\sqrt{3}+3\right)}$ C) $\dfrac{13\left(2\sqrt{3}-3\right)\pi}{8}$ D) $\dfrac{\pi}{8\left(2\sqrt{3}+3\right)}$

57. Let $g:R\to R$ be a non constant twice differentiable such that $g'\left(\dfrac{1}{2}\right)=g'\left(\dfrac{3}{2}\right)$. If a real valued function f is defined as $f(x)=\dfrac{1}{2}\left[g(x)+g(2-x)\right]$, then

A) $f''(x)=0$ for atleast two x in $(0,2)$

B) $f''(x)=0$ for exactly one x in $(0,1)$

C) $f''(x)=0$ for no x in $(0,1)$

D) $f'\left(\dfrac{3}{2}\right)+f'\left(\dfrac{1}{2}\right)=1$

58. The area (in square units) of the region bounded by the parabola $y^2=4(x-2)$ and the line $y=2x-8$ is

A) 8 B) 9 C) 6 D) 7

59. Let $y=y(x)$ be the solution of the differential equation $\sec x\,dy+\left\{2(1-x)\tan x+x(2-x)\right\}dx=0$ such that $y(0)=2$. Then $y(2)$ is equal to

A) 2 B) $2\{1-\sin(2)\}$ C) $2\{\sin(2)+1\}$ D) 1

60. Let (α,β,γ) be the foot of perpendicular from the point $(1,2,3)$ on the line $\dfrac{x+3}{5}=\dfrac{y-1}{2}=\dfrac{z+4}{3}$ then $19(\alpha+\beta+\gamma)$ is equal to

A) 102 B) 101 C) 99 D) 100

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61. Two integers x and y are chosen with replacement from the set $\{0,1,2,3,.....,10\}$. Then the probability that $|x-y|>5$ is

A) $\frac{30}{121}$ B) $\frac{62}{121}$ C) $\frac{60}{121}$ D) $\frac{31}{121}$

62. If the domain of the function $f(x)=\cos^{-1}\left(\frac{2-|x|}{4}\right)+\left(\log_e(3-x)\right)^{-1}$ is $[-\alpha,\beta)-\{\gamma\}$, then $\alpha+\beta+\gamma$ is equal to:

A) 12 B) 9 C) 11 D) 8

63. Consider the system of linear equations $x+y+z=4\mu,\ x+2y+2\lambda z=10\mu,\ x+3y+4\lambda^2 z=\mu^2+15$, where $\lambda,\ \mu\in R$. Which one of the following statements is NOT correct?

A) The system has unique solution if $\lambda\neq\frac{1}{2}$ and $\mu\neq 1,15$

B) The system is inconsistent if $\lambda=\frac{1}{2}$ and $\mu\neq 1$

C) The system has infinite number of solutions if $\lambda=\frac{1}{2}$ and $\mu=15$

D) The system is consistent if $\lambda\neq\frac{1}{2}$

64. If the circles $(x+1)^2+(y+2)^2=r^2$ and $x^2+y^2-4x-4y+4=0$ intersect at exactly two distinct points, then

A) $5<r<9$ B) $0<r<7$ C) $3<r<7$ D) $\frac{1}{2}<r<7$

65. If the length of the minor axis of an ellipse is equal to half of the distance between the foci, then the eccentricity of the ellipse is:

A) $\frac{\sqrt{5}}{3}$ B) $\frac{\sqrt{3}}{2}$ C) $\frac{1}{\sqrt{3}}$ D) $\frac{2}{\sqrt{5}}$

66. Let M denote the median of the following frequency distribution.

Class	0-4	4-8	8-12	12-16	16-20
Frequency	3	9	10	8	6

Then $20M$ is equal to:

A) 416 B) 104 C) 52 D) 208

67. If $f(x)=\begin{vmatrix} 2\cos^4 x & 2\sin^4 x & 3+\sin^2 2x \\ 3+2\cos^4 x & 2\sin^4 x & \sin^2 2x \\ 2\cos^4 x & 3+2\sin^4 x & \sin^2 2x \end{vmatrix}$ then $\frac{1}{5}f'(0)$ is equal to ____

A) 0 B) 1 C) 2 D) 6

68. Let $A(2,3,5)$ and $C(-3,4,-2)$ be opposite vertices of a parallelogram $ABCD$. If the diagonal $\overrightarrow{BD}=\hat{i}+2\hat{j}+3\hat{k}$ then the area of the parallelogram is equal to

A) $\frac{1}{2}\sqrt{410}$ B) $\frac{1}{2}\sqrt{474}$ C) $\frac{1}{2}\sqrt{586}$ D) $\frac{1}{2}\sqrt{306}$

69. If $2\sin^3 x+\sin 2x\cos x+4\sin x-4=0$ has exactly 3 solutions in the interval $\left[0,\frac{n\pi}{2}\right], n\in N$, then the roots of the equation $x^2+nx+(n-3)=0$ belong to:

A) $(0,\infty)$ B) $(-\infty,0)$ C) $\left(-\frac{\sqrt{17}}{2},\frac{\sqrt{17}}{2}\right)$ D) Z

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70. Let $f:\left[-\frac{\pi}{2},\frac{\pi}{2}\right]\to R$ be a differentiable function such that $f(0)=\frac{1}{2}$. If the $\lim\limits_{x\to 0}\frac{\int_0^x f(t)dt}{e^{x^2}-1}=\alpha$, then $8\alpha^2$ is equal to:

A) 16 B) 2 C) 1 D) 4

SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

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71. Let the latus rectum of the hyperbola $\frac{x^2}{9}-\frac{y^2}{b^2}=1$ subtend an angle of $\frac{\pi}{3}$ at the centre of the hyperbola. If b^2 is equal to $\frac{l}{m}\left(1+\sqrt{n}\right)$, where l and m are co-prime numbers, then $l^2+m^2+n^2$ is equal to _____

72. Let $A=\{1,2,3,.....,7\}$ and let $P(A)$ denote the power set of A. If the number of functions $f:A\to P(A)$ such that $a\in f(a),\forall a\in A$ is m^n, m and $n\in N$ and m is least, then $m+n$ is equal to

73. The value $9\int_0^9\left[\sqrt{\frac{10x}{x+1}}\right]dx$, where $[t]$ denotes the greatest integer les than or equal to t, is _____

74. Number of integral terms in the expansion of $\left\{7^{\left(\frac{1}{2}\right)}+11^{\left(\frac{1}{6}\right)}\right\}^{824}$ is equal to _____

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75. If the function $f(x)=\begin{cases}\dfrac{1}{|x|}, & |x|\geq 2\\ ax^2+2b, & |x|<2\end{cases}$ is differentiable on R, then $48(a+b)$ is equal to _____.

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